Generator-induced invariants in lattice public keys: a pre-standardization rank diagnostic

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October 2026

Abstract

A public-key construction that builds its public object by applying a fixed family of linear maps to random seeds—to compress the key, to inherit a symmetry, or to support a reduction—produces an object that is the image of a single fixed *design matrix*. The row space of that matrix, and with it its rank, is then imposed on the public object whatever the seeds. When the rank ρ of the design matrix is below the ambient dimension, every public key is rank-deficient: exact linear algebra distinguishes it from the uniform matrix a security reduction assumes, and recovering the noise reduces to an LWE instance of dimension at most ρ. The argument takes one line—the rank of a product is at most the rank of either factor—yet the weakness is easy to miss: a rank check on a realized instance of dimension n can certify only $\rho \geq n$, so a check run below the ceiling returns a clean result on a broken scheme. We make this concrete with a reproducible case study, a withdrawn sedenion Module-LWE proposal whose public matrix has rank at most 76 at every module rank and whose own structural check, run at module rank 4, could not see the ceiling. We package the lesson as an exact and cheap diagnostic—count, then compute the rank of the design matrix over the working field at the deployed size, and confirm it on an instance larger than the ceiling—with a dependency-free implementation that reproduces the case study and passes two structured, sound controls, and we place the mechanism among published breaks in which structure introduced for a benefit became the attack surface. Passing the diagnostic is necessary for any security argument that assumes a uniform, full-rank matrix, but not sufficient, and it costs less than the lattice estimate it should precede.

Epistemic conventions. As in the author's other work, claims are tiered. Every proposition and corollary below is T1: proven from the stated definitions, or verified by exact computation. The remarks are held to the same standard: they state only what follows from the propositions, from the cited literature, or from exact computation that the accompanying self-test checks. Section 5 classifies published breaks; that classification is interpretive, and where it reaches beyond Proposition 1 it says so.

1 Introduction

Uniformly random public keys are large, and designers routinely replace them with a *generated* object that is cheaper to store or transmit: the cyclic and negacyclic structure of Ring- and Module-LWE, the quasi-cyclic and dyadic codes of compact code-based schemes, group-action images, and orbit constructions of many kinds. The generator is always introduced for a reason: a smaller key, a symmetry the designer believes is benign or even helpful, or a structural argument that the construction reduces to a well-studied problem.

The thesis of this note is that such a generator is a fixed linear gadget, and that random seeds fed through a fixed linear gadget cannot produce anything outside its image. If that image is confined—a rank ceiling, equivalently a kernel common to every key—the public object inherits

the confinement however much entropy the seeds carry. When the ceiling is below the ambient dimension, the consequences are immediate: the public key is distinguishable from the uniform matrix a reduction assumes, and recovery drops from the ambient dimension to at most the rank of the design matrix. Not every invariant of a generator is a weakness: the multiplication maps of Ring- and Module-LWE share fixed subspaces and are sound, and Remark 1 says what separates the two.

None of the linear algebra is deep, and that is the point. The mechanism is elementary, and the diagnostic that catches it takes seconds of exact arithmetic. Yet the weakness is easy to ship, because a rank check on a realized instance certifies only that the ceiling is at least the size of that instance. Below the ceiling, a generated object can be full rank and indistinguishable from a uniform control; a check run there—the diligent thing to do—returns a clean result. A scheme can therefore pass its own structural test and still carry a fatal ceiling, if the test was run on too small an instance.

Contributions.

1. A general statement (Section 2): a seed-generated public object factors as $M = R \cdot D$ with D a fixed design matrix, so $\operatorname{rank} M \leq \operatorname{rank} D$ for every instance size and every draw of the seeds. The same factorization gives a kernel shared by every public key (a distinguisher) and confines the LWE error to an affine subspace of dimension at most $\operatorname{rank} D$ (a reduced recovery problem), and two counts bound $\operatorname{rank} D$ before anything is computed.

2. A visibility principle (Proposition 2): a full-rank check at ambient dimension n certifies only $\operatorname{rank} D \geq n$, and a deficiency is guaranteed to show only once $n > \operatorname{rank} D$. Computing $\operatorname{rank} D$ from the design certifies every size at once.

3. A reproducible case study (Section 3): a withdrawn sedenion Module-LWE proposal in which exactly this mechanism gives a ceiling of 76 at every module rank, with an account of why the scheme's own check, run at module rank 4, could not see it.

4. An exact, cheap diagnostic (Section 4) with a dependency-free implementation that reproduces the case study as a regression test, takes its verdict at the deployed size, and passes two structured, sound controls: a sedenion module matrix with independent uniform entries and an ML-KEM-shaped module lattice.

5. A short classification of published breaks of the same shape (Section 5), with an account (Section 6) of what the theorem does and does not cover.

Scope. This is a defensive, methodological note. The case study is the author's own withdrawn proposal, and the ML-KEM design [21] appears only as a correctly designed control, which the diagnostic passes. No attack on any deployed or standardized system is given or implied.

2 The mechanism

2.1 Setup

Let $\mathbb{F}$ be a field. A *seed-generated public object* is built as follows. The scheme fixes a finite family of $\mathbb{F}$-linear maps $T_0, \ldots, T_{r-1} \colon \mathbb{F}^d \to \mathbb{F}^d$ (the *generator*), draws independent random seeds, applies the generator to the seeds, and combines the results $\mathbb{F}$-linearly. A *seed* here is the vector of coefficients the generator consumes, after any nonlinear expansion: ML-KEM derives its matrix from one 32-byte string, but its generator consumes $256k^2$ coefficients [21].

Write M for the matrix of the $\mathbb{F}$-linear map the scheme exposes—for an LWE-type key, the map $s \mapsto As$. Because the assembly is linear in the seeds and linear in the fixed generator images, M is bilinear in the two and factors as

$$M \;=\; R \cdot D,$$

where $R \in \mathbb{F}^{a\times g}$ carries all of the seed randomness (its entries are linear in the seed coordinates) and $D \in \mathbb{F}^{g\times b}$ is *fixed*: its entries are determined by the generator and the combination rule and do not depend on the seeds. We call D the *design matrix*. Only its row space matters below, so D may be replaced by any matrix with the same row space, for instance with repeated rows removed. A uniform object factors trivially, with D the identity.

2.2 The rank ceiling

Proposition 1 (rank ceiling)**.** *For every instance and every draw of the seeds,* $\operatorname{rank} M \le \operatorname{rank} D$.

Proof. $M = RD$ and $\operatorname{row}(RD) \subseteq \operatorname{row}(D)$, so $\operatorname{rank} M \le \operatorname{rank} D$. □

The content lies entirely in the fact that D is fixed. Seed entropy moves M within the row space of D; it cannot raise the rank above $\operatorname{rank} D$. If $\operatorname{rank} D$ is below the ambient dimension, every public key the scheme can emit is rank-deficient, with probability one.

Two counts bound $\operatorname{rank} D$ before anything is computed: the number of distinct rows of D and the number of distinct columns. When each block row of M reuses one seed of length ℓ through the generator, as in the case study, the design needs only ℓd rows for $d \times d$ blocks, whatever the generator; and D has no more distinct columns than there are distinct generator images (Section 2.3). If either count is below the ambient dimension at the deployed size, the scheme fails without further computation.

Corollary 1 (distinguisher)**.** *Let* $\rho = \operatorname{rank} D < b$. *Then* $\ker D \subseteq \ker M$ *for every public key, and* $\dim\ker D = b - \rho$. *In particular, a fixed nonzero* $x \in \ker D$*—computable from the design alone—satisfies* $Mx = 0$ *for every public key. A uniform* $M \in \mathbb{F}_q^{a\times b}$ *does so with probability* q^{-a}*, and so does a module matrix whose entries are independent uniform elements of a* d*-dimensional algebra, provided right multiplication by some block* x_j *of* x *is invertible.*

Proof. If $Dx = 0$ then $Mx = R(Dx) = 0$; the dimension count is rank–nullity. For uniform M, each of the a rows is orthogonal to $x \neq 0$ independently with probability $1/q$. For independent uniform entries A_{ij}, block row i of Mx is $\sum_j A_{ij}x_j$; the term $A_{ij}x_j$ is uniform when right multiplication by x_j is invertible, so the block row vanishes with probability q^{-d}, independently over the a/d block rows. □

A structured matrix is distinguishable from a uniform $M \in \mathbb{F}_q^{a\times b}$ whether or not it is weak—ML-KEM's is, through its negacyclic blocks—so the comparison that matters is with the distribution the scheme's security argument assumes. The corollary covers both cases; Section 3.2 applies it to the case study.

Corollary 2 (recovery)**.** *Write the public key in LWE form* $t = Ms + e$ *with* $M \in \mathbb{F}_q^{a\times b}$*, and let* $V = \operatorname{col}(M)$ *and* $\rho_M = \dim V \le \operatorname{rank} D$. *Then* $t - e \in V$*, so* e *lies in the known affine subspace* $t + V$ *of dimension* ρ_M. *Choosing* ρ_M *coordinates* I *on which a basis of* V *is invertible expresses the remaining* $a - \rho_M$ *coordinates of* e *as fixed affine functions of* e_I*: an LWE instance with secret* e_I *of dimension* ρ_M *and* $a - \rho_M$ *samples. Equivalently, a parity-check matrix* L *for* V*, with* $LM = 0$*, gives* $Lt = Le$.

Proof. Every $v \in V$ is determined by v_I, with $v_{\bar I} = Cv_I$ for a matrix C computable from M. Applied to $v = t - e$ this gives $e_{\bar I} = Ce_I + (t_{\bar I} - Ct_I)$, in which e_I and $e_{\bar I}$ are short. □

From noise to message. Recovering the noise suffices to decrypt whenever decryption is linear. Suppose a ciphertext is $u = M^*r + e_1$ and $v = \langle t, r\rangle + e_2 + \mathrm{encode}(m)$, where M^* is the adjoint of M for the pairing used in decryption and r, e_1, e_2 are short. Since $\operatorname{rank} M^* = \operatorname{rank} M$, the argument of Corollary 2 applied to u recovers e_1 as it recovered e. Any s' with $Ms' = t - e$ then gives $\langle s', u - e_1\rangle = \langle Ms', r\rangle = \langle t - e, r\rangle$, so $v - \langle s', u - e_1\rangle = \langle e, r\rangle + e_2 + \mathrm{encode}(m)$, a small perturbation of the encoded message.

Remark 1. Rank is the cheapest invariant to read off, and the only one this argument transfers: the factorization gives $\mathrm{row}(M) \subseteq \mathrm{row}(D)$, equivalently $\ker D \subseteq \ker M$, and nothing more. A subspace that every key preserves is a different matter, because it need not be a weakness. Over $\mathbb{F}_{3329}$ the polynomial $X^{256} + 1$ splits into 128 quadratic factors, so every multiplication map of ML-KEM's ring preserves the same 128 two-dimensional subspaces (its NTT slots); over $\mathbb{F}_{8380417}$ it splits into linear factors, so the multiplication maps of ML-DSA share 256 eigenvectors [21]. Projecting onto such a subspace does shrink the secret, yet both schemes are sound, because the projected noise is close to uniform. An invariant subspace is a weakness only when the projection keeps the noise short, as the restriction to coordinates in Corollary 2 does, and as evaluation at a suitable root of the defining polynomial (for instance 1) does in the provably weak instances of Ring-LWE [12, 8]; Peikert [22] shows that those attacks do not apply when the error is wide enough relative to the ring's geometry. A test for invariant subspaces must check that condition, not the subspace alone, and this note's diagnostic does not attempt it.

2.3 Why the ceiling hides

The design matrix of an instance is assembled from the generator images that instance uses. Suppose the generator contributes c independent images in total; for a cyclic generator of order r acting on a d-dimensional alphabet, $c \le rd$, and the column space of the design stops growing once an instance uses all r powers. Let D be the *full design*, which contains every generator image, and let $\rho = \operatorname{rank} D$. Let $M_k = R_k D_k$ be the realized public map at instance size k, and assume that D_k is assembled from columns of D, so that $\mathrm{col}(D_k) \subseteq \mathrm{col}(D)$ and $\operatorname{rank} D_k \le \rho$; this holds whenever the generator has finitely many images. Let $n(k)$ be the smaller dimension of M_k, which is the rank a uniform matrix of that shape has with high probability.

Proposition 2 (visibility)**.**

(i) If $\operatorname{rank} M_k = n(k)$*, then* $\rho \ge n(k)$*, and Proposition 1 excludes no larger value of* ρ*: a full-rank result at size* k *is consistent with every ceiling* $\rho \ge n(k)$*, including ceilings far below the ambient dimension* $n(k')$ *of a larger deployed size* k'*.*

(ii) If $n(k) > \rho$*, then* $\operatorname{rank} M_k \le \rho < n(k)$*: the deficiency is visible.*

Proof. Proposition 1 applied to $M_k = R_k D_k$, together with $\operatorname{rank} D_k \le \rho$ and the shape of M_k, gives $\operatorname{rank} M_k \le \min(n(k), \rho)$. If $\operatorname{rank} M_k = n(k)$, this forces $n(k) \le \rho$ and constrains ρ no further. Table 1 exhibits a design with $\rho = 76$ whose realizations have full rank at every $k \le 4$ ($n(k) \le 64$), although the deployed size has $n(k') = 1024$. Part (ii) is immediate. □

Remark 2. In the case study of Section 3, the realized rank equals $\operatorname{rank} N_k$, the rank of the design restricted to the powers $\sigma^0, \ldots, \sigma^{k-1}$ that an instance of size k uses, at every size tested (Table 1): $16k$ for $k \le 4$, then 72 and 74 at $k = 5$ and 6, and $\rho = 76$ from $k = 7$, once all seven powers appear ($\sigma^7 = \mathrm{id}$). Proposition 1 is therefore tight here for generic seeds. The deficiency first appears at $k = 5$, the smallest k with $n(k) > \rho$, and a check at $k = 4$ is precisely the uninformative case (i). This is a property of the case-study design, established by exact computation and checked by the self-test; it is not a theorem about all designs.

Two practical rules follow. An empirical rank check certifies only the size it is run at, so a check below the deployed size certifies nothing about the deployed key. And computing $\rho = \operatorname{rank} D$ from the design certifies every size at once: it is cheaper and stronger than any empirical check, which then serves only to confirm that the factorization was done correctly.

3 Case study: a sedenion Module-LWE proposal

We instantiate the mechanism on a concrete, withdrawn scheme. The numerical claims in this section are those of the negative-result note that broke it [16], to which we refer for the computations; the aim here is to exhibit the general mechanism, not to re-derive the break. The scheme and the break are the author's own.

3.1 The construction and its intent

The scheme, SLWE [15], is a Module-LWE key encapsulation mechanism [23, 19] over the 16-dimensional sedenion algebra $\mathbb{S}_q = \mathbb{S} \otimes \mathbb{F}_q$, the fourth Cayley–Dickson algebra. Its public matrix $A \in \mathbb{S}_q^{k \times k}$ is built from orbits of an order-7 coordinate permutation σ—a "Singer cycle" inherited from the symmetry of the Fano plane—as $A_{ij} = \sigma^j(\mathrm{seed}_i)$, with independent uniform seeds $\mathrm{seed}_i \in \mathbb{S}_q$. The orbit structure was deliberate: it compresses the key (each row comes from one seed) and was meant to preserve the algebra's zero-divisor interface. Non-associativity was argued to remove an ideal-lattice attack class and so to reduce security to that of a generic module lattice. Two parameter sets were published with a claim of NIST Category 5: $n = 512$ ($k = 32$) with $q \approx 2^{32}$, and $k = 64$ ($n = 1024$) with $q = 911$.

3.2 The factorization and the ceiling

Let $M \in \mathbb{F}_q^{16k \times 16k}$ be the matrix of $s \mapsto As$, so that $M_{16i+u,\,16j+c}$ is coefficient u of $\sigma^j(\mathrm{seed}_i)\, e_c$. Concretely, σ fixes e_0 and e_8, maps $e_t \mapsto e_{t+1}$ for $1 \le t \le 6$ and $e_7 \mapsto e_1$, and acts in the same way on $e_9, \ldots, e_{15}$; the product is the Cayley–Dickson doubling $(a, b)(c, d) = (ac - \bar{d}b,\ da + b\bar{c})$. Because σ is linear and the sedenion product is bilinear, M factors exactly as in Section 2:

$$M = (S \otimes I_{16})\, N_k,$$

where $S \in \mathbb{F}_q^{k \times 16}$ holds the seeds (this is R) and N_k is assembled from the columns of a fixed integer matrix $N \in \mathbb{Z}^{256 \times 112}$ whose entry $N_{16m+u,\,16j+c}$ is coefficient u of $\sigma^j(e_m)\, e_c$ (this is D). Since $\sigma^7 = \mathrm{id}$, the column blocks of N_k repeat with period 7, so $\mathrm{col}(N_k) \subseteq \mathrm{col}(N)$, with equality once $k \ge 7$.

Counting alone already bounds the rank. The design N has 256 rows—16 seed coordinates times 16 rows per block—and 112 columns—seven powers of σ times 16—so $\operatorname{rank} M \le 112$ at every k, and for $k > 7$ the repetition $A_{i,j+7} = A_{ij}$ is visible in the deployed matrix by inspection. Exact computation sharpens 112 to 76: $\operatorname{rank} N = 76$ over $\mathbb{Q}$ and modulo 911 and $q = 4\,294\,977\,961$ [16]. Because N has integer entries, its rank modulo any prime is at most its rank over $\mathbb{Q}$, so the ceiling of 76 holds at every modulus at once. Over $\mathbb{F}_{911}$, which contains the seventh roots of unity, a discrete Fourier transform over the power index j splits $\mathrm{col}(N)$ into a part of dimension 16 for the trivial character and six parts of dimension 10, so $76 = 16{+}6{\cdot}10$ [16]. The other common Cayley–Dickson convention, $(a, b)(c, d) = (ac - d\bar{b},\ \bar{a}d + cb)$, gives the same ranks. The accompanying diagnostic re-derives $\operatorname{rank} N = 76$ over $\mathbb{Q}$ by fraction-free elimination and modulo 911, 2731, $2^{31} - 1$ and q. By Proposition 1, $\operatorname{rank} M \le 76$ at every module rank, against $16k$ for a matrix with uniform entries. Table 1 lists the exact ranks.

By Corollary 2, recovering the noise is an LWE instance of dimension 76 with $16k - 76$ samples. The lattice estimator [2, 3] places it at $2^{38.6}$ operations at the first parameter set (436 samples) and at $2^{40.4}$ at the second (948 samples), with the block size at the floor of its

k	1	2	3	4	5	6	7	8	16	32	64
$n(k) = 16k$	16	32	48	64	80	96	112	128	256	512	1024
rank M_k (Singer orbit)	16	32	48	64	**72**	**74**	**76**	**76**	**76**	**76**	**76**
rank M_k (uniform entries)	16	32	48	64	80	96	112	128	256	512	1024

Table 1: Exact rank of the flattened public map at module rank k: Singer orbit versus a matrix with independent uniform sedenion entries, the distribution the scheme's reduction assumed (a uniform matrix over $\mathbb{F}_q$ has the same ranks). The values over $\mathbb{F}_{911}$ and over $\mathbb{F}_q$ with $q = 4\,294\,977\,961$ are identical; the Singer-orbit row is as reported in [16]. The self-test of the accompanying diagnostic (Section 4) regenerates the columns $k \leq 8$ over $\mathbb{F}_{911}$ and checks that each Singer-orbit entry equals rank N_k. Bold entries fall short of $n(k)$. The columns $k \leq 4$ are identical for the two matrices: a check there cannot see the ceiling (Proposition 2(i)).

model, so that reduction at the scale of LLL suffices [16]. The encryption map is the conjugate transpose of A [16], of the same rank, so each ciphertext's noise falls in the same way, and the argument after Corollary 2 then yields the message; [16] recovers the noise and the encrypted bit exactly at the first parameter set. Distinguishing the public matrix from the distribution its reduction assumed is cheaper still: by Corollary 1, it takes Gaussian elimination. A fixed $x \in \ker N_8$ is annihilated by every Singer-orbit key at $k = 8$, while right multiplication by one of its blocks is invertible, so a matrix with independent uniform sedenion entries annihilates it only with probability q^{-128}; the self-test checks both facts over $\mathbb{F}_{911}$.

3.3 Why the scheme's own check missed it

The proposal included a structural self-test: a five-test Gram–Schmidt profile of the lattice, looking for exactly the kind of anomaly at issue. It reported no rank deficiency, no anomalous short vectors and no invariant sublattice relative to a uniform matrix [15]. It was run on a 64×64 proxy at module rank $k = 4$, and the proposal itself labelled it a toy-scale result that "does not establish anything at production dimension".

Proposition 2 makes that caveat exact. At $k = 4$ the public map has $n(4) = 64$ columns, below the ceiling of 76, so by part (i) a full-rank result certified only $\rho \geq 64$. An instance of size 4 also uses only four of the seven powers of σ, so even the repetition $A_{i,j+7} = A_{ij}$ of the deployed matrix was absent from it. The deployed matrix at $k = 64$ has $n = 1024$; every value $64 \leq \rho < 1024$ was left open, and the truth was $\rho = 76$. The check returned a clean result not because the scheme was sound but because it was run below the ceiling. Table 1 shows the trap at a glance: the columns $k \leq 4$ are identical for the Singer-orbit and the uniform matrix, and they separate only from $k = 5$.

3.4 Non-associativity was beside the point

The scheme's security narrative rested on the non-associativity of the sedenions and on the σ-symmetry of A. Neither bears on the break. Decryption uses the bilinear conjugate-norm pairing, so every map the attacker needs is $\mathbb{F}_q$-linear, and the argument of Section 2 uses only linearity. The permutation σ is not an automorphism of the sedenion product but a relabelling of coordinates: it breaks $\sigma(xy) = \sigma(x)\sigma(y)$ on 144 of the 256 products of basis elements [16]. It is not even a collineation of the Fano plane as the product labels it, whose lines are $\{1,2,3\}$, $\{1,4,5\}$, $\{1,6,7\}$, $\{2,4,6\}$, $\{2,5,7\}$, $\{3,4,7\}$ and $\{3,5,6\}$: it sends $\{1,2,3\}$ to $\{2,3,4\}$. So the "symmetry" of A is not an algebraic symmetry at all. The lesson travels: advertised algebraic richness can be orthogonal to the property that carries security, and a security argument phrased in terms of the rich structure may be answering the wrong question.

3.5 The repair that was not one

Once the deficiency was found, the natural repair was to replace the Singer-orbit matrix with a uniform one, which removes the ceiling. It does not rescue the claim: at the first parameter set the modulus-to-noise ratio alone puts the cheapest attack near 2^{51}, and at the second near 2^{167}, about 95 bits below the 2^{262} the same estimator gives ML-KEM-1024 [16]. Clearing the structural flag one has found does not clear a parameter problem one has not. Passing the rank test of Section 4 is necessary for a uniform-matrix security argument, never sufficient.

4 The diagnostic

The mechanism gives a checklist that a designer or evaluator can run before trusting a compressed or structured public key, at a cost far below the lattice estimate it should precede.

0. **Count.** Count the distinct rows and the distinct columns of the design at the deployed size; $\operatorname{rank} D$ is at most the smaller. If that is below the ambient dimension, the scheme fails before any elimination (in the case study, 112 against 512 and 1024).

1. **Factor.** Write the construction as $M = R \cdot D$, separating the seed randomness R from the fixed generator gadget D. For a uniform object the factorization is trivial, with D the identity, and the test passes.

2. **Build the design** at each deployed size and, if the generator has finitely many images, the full design that contains them all (for a cyclic generator of order r, all r powers), which bounds every size at once.

3. **Compute the ceiling exactly.** Compute $\rho = \operatorname{rank} D$ over the working field $\mathbb{F}_q$ with exact arithmetic, not floating point. When D has integer entries, also compute its rank over $\mathbb{Q}$: the rank modulo any prime is at most that, so it bounds every modulus at once (Proposition 1).

4. **Confirm on a large enough instance.** Realize M at the smallest size with $n(k) > \operatorname{rank} D_k$ and compare its rank, over several seed sets, with matrices of the same shape drawn from the distribution the scheme's security argument assumes. A check below that size is uninformative (Proposition 2); a realized rank above $\operatorname{rank} D_k$ contradicts Proposition 1 and means the factorization is wrong.

5. **Flag.** Report a ceiling if ρ is below the ambient dimension at the deployed size. It gives a distinguisher (Corollary 1) and a recovery problem of dimension at most ρ (Corollary 2), at which the lattice estimate must be rerun. Invariant subspaces are not flagged: they are a weakness only under the condition of Remark 1, which this test does not check.

The test is exact linear algebra on a matrix the size of the design: it takes seconds, needs no lattice reduction or estimator, and costs less than the Core-SVP estimate a candidate already runs. It belongs *before* that estimate, because a positive result changes the dimension at which the estimate must be run and, when ρ is as small as in the case study, makes it moot.

Where to look. The factorization of step 1 exists whenever the public object is a generator applied to seeds: cyclic or orbit-generated matrices, images under linear group actions, structured-code substitutions, and any object described as compressed, symmetric, or inheriting an algebraic structure.

Implementation. A standalone implementation in pure Python, with no dependencies, accompanies this note (`design_matrix_rank.py`; Section 7). A user describes a construction by implementing four methods—`design_matrix(k)`, `realize()`, `ambient_rank()` and `random_seeds()`—and listing its parameter sets, each a deployed size with its working modulus; optional methods supply the full design and a sampler for the distribution the security argument assumes. On the designs the user supplies (steps 1 and 2), the tool runs steps 0, 3, 4 and 5, takes its verdict at each deployed size, and reports an error if a realized rank exceeds $\operatorname{rank} D_k$. It ships with the case study of Section 3 (verdict *fail*: ceiling 76 at both parameter sets, and 112 by counting alone) and two structured, sound controls (verdict *pass*): a sedenion module matrix with independent uniform entries, and an ML-KEM-shaped module of independent uniform elements of $\mathbb{Z}_{3329}[X]/(X^{256}+1)$ at $k = 2, 3, 4$. A `--self-test` mode reproduces the columns $k \leq 8$ of Table 1, $\operatorname{rank} N = 76$ and the other facts this note computes, checks the verdicts on the case study, on the controls and on designs built to fail, and exits with an error on any mismatch, so it doubles as a regression test pinned to the published result. The empirical curve it prints marks every size below the ceiling as one at which the collapse is invisible: Section 3.3 in executable form.

5 The pattern recurs

The case study is one instance of a recurring shape: *a structural feature introduced for a benefit carries an invariant, or an auxiliary datum, that an attacker reads off.* Only the first item below is an instance of Proposition 1. The others are published breaks, cited as established results; what we add is the classification—that each turns a structure introduced for a benefit into the lever—which is interpretive, not a theorem. All are breaks of proposed or withdrawn constructions or instantiations.

- **A rank ceiling from a generator.** The case study of Section 3. Benefit sought: a compressed key and an inherited symmetry. Lever: the rank of the design matrix. This is the linear-image case that Proposition 1 governs directly.
- **Structured codes in McEliece.** Random binary Goppa codes give a sound McEliece instance, but substituting a structured code to shrink the key has failed repeatedly: generalized Reed–Solomon codes [25], compact quasi-cyclic and dyadic variants [14], and algebraic-geometry codes [9], each broken by a structural attack that recovers the hidden code from the public generator matrix. Dimension counts are a common tool, as in Section 4: the attack on algebraic-geometry codes works with componentwise products of the public code, whose dimensions are abnormally small for such codes [9], and even binary Goppa codes become distinguishable from random codes at high rate, where a linear system built from the public key has an abnormally large solution space [13]. Structure added for size became the lever.
- **Layers in multivariate signatures.** Rainbow, a finalist in the third round of NIST's process, layered Unbalanced Oil and Vinegar for efficiency. The layers make some linear combinations of the public quadratic forms abnormally low in rank, the lever of MinRank attacks; Beullens' rectangular MinRank attack [4] and his key recovery, which takes about a weekend on a laptop at the level-I parameters [5], exploit exactly that structure.
- **Overstretched NTRU.** With a large modulus at fixed noise, a dense sublattice appears, which subfield and sublattice attacks exploit [1, 18]; the onset was later located as the "fatigue point" [11]. Here the structure comes from a parameter choice made for functionality rather than from an explicit generator, but the lever is again a fixed low-dimensional structure in the public object. It is also the published counterpart of Section 3.5: there too a modulus too large for its noise, not a generator, was the lever.

- **Principal ideals and short generators.** Compact key constructions over principal ideals, such as Soliloquy [6], admit short-generator recovery [10]: the ideal structure chosen for compactness is the route to the key.

- **Weak instances of Ring-LWE.** Every multiplication map of a Ring-LWE ring preserves the same subspaces. For some defining polynomials and error distributions, projecting onto one of them keeps the error short and breaks the instance [12, 8]; with errors wide enough relative to the ring's geometry the projection is harmless [22]. This is the invariant-subspace case of Remark 1, together with the condition it needs.

- **SIDH and SIKE: auxiliary data for functionality (an analogy).** The images $\phi(P), \phi(Q)$ of torsion points are published so that the key exchange can complete. The 2022 attacks, all published at EUROCRYPT 2023 [7, 20, 24], use Kani's theorem [17] to embed the secret isogeny in a higher-dimensional abelian variety and turn exactly that auxiliary data into key recovery, in polynomial time in full generality [24]. This is not a rank ceiling and not an instance of Proposition 1; it marks the boundary of the linear-image argument while sharing its moral: the structure published to make the scheme work was the attack surface.

The question this classification recommends asking of any structured proposal is simple: *what did the structure I added give the attacker?* For the linear-image cases, Section 4 answers it mechanically and cheaply; for the others it is a design discipline rather than a single test.

6 Limitations

- **Linear-image generators only.** Proposition 1 is exact and general for public objects bilinear in the seeds and a fixed generator. Nonlinear generators, and the SIDH case of Section 5, lie outside it; there the comparison is an analogy, and Section 5 says so.

- **Necessary for a uniform-matrix argument, not sufficient.** A full-rank design shows only that this class of weakness is absent. Invariant subspaces, which every ring-structured design has, are a weakness only under the condition of Remark 1, which the test does not check. Section 3.5 is the cautionary case: with the structure removed, the scheme still failed on its modulus-to-noise ratio, which this test does not judge.

- **Structure is not the enemy.** Module-LWE and binary-Goppa McEliece are structured and sound, and the ML-KEM-shaped control passes the diagnostic. The claim is narrow and testable: generator-induced invariants must be computed, over the working field and for the design at the deployed size, before a structured public object is trusted.

- **Dimension, not cost.** Corollary 2 gives the dimension to which recovery reduces; the concrete cost is a lattice-estimator question that depends, as Section 3.5 shows, on parameters the rank test does not see. A ceiling close to the ambient dimension may leave the reduced problem hard; the case study is fatal because $\rho = 76$.

7 Reproducibility

The diagnostic is a single file in the public repository `gifgaf0/gifgaf0.github.io`, and a copy accompanies this note on Zenodo:

file	`generator_induced_invariants/diagnostics/design_matrix_rank.py`
SHA-256	`b6a368c9b9f1e269b9dca93bc0d5f8c81ab5f5797687352fe3c4154e147f9818`
self-test	`python3 design_matrix_rank.py --self-test`

It runs under Python 3.8 or later with no other dependencies. The self-test regenerates the columns $k \le 8$ of Table 1 over $\mathbb{F}_{911}$ and checks that they equal rank N_k; it re-derives rank $N = 76$ over $\mathbb{Q}$ by fraction-free elimination and modulo 911, 2731, $2^{31} - 1$ and $q = 4\,294\,977\,961$, and over $\mathbb{Q}$ under the other Cayley–Dickson convention; it checks the count of 112, the split $76 = 16 + 6 \cdot 10$, the 144 failing basis products, the image of $\{1, 2, 3\}$ under σ and the kernel test of Section 3.2; and it checks the verdicts on the case study, on both controls, on two designs built to fail, and on one whose design does not factor its realization. The full break of the case-study scheme—both independent rank implementations, one written blind from a textual specification, the exact ranks computed with python-flint, and the lattice-estimator runs—is archived with [16].

Acknowledgements. AI agents (Anthropic's Claude) drafted this note, generalized its mechanism from the case study, and wrote and ran the accompanying diagnostic; a review by Google's Gemini suggested the seed-length count of Section 2. The author directed the work, reviewed the text, and takes responsibility for it.

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